

Final Assignment - AB testing in SQL

AB Testing for Item Orders and Views

Question 1: Compare the `final_assignments_qa` table to the assignment events we captured for `user_level_testing`. Write an answer to the following question: Does this table have everything you need to compute metrics like 30-day view-binary?

1. Data Quality Check

	item_id	test_a	test_b	test_c	test_d	test_e	test_f
1	2512	1	0	1	1	0	1
2	482	0	1	1	1	0	0
3	2446	0	1	1	0	1	0
4	1312	0	0	0	0	0	1
...	...	...	...	...	...	...	...

This table only shows the first 1,000 rows.

The four columns needed for a binary AB test are:

- item identifier (`item_id`)
- test assignment status (`test_a` through `test_f`)
- test number (`test_a` through `test_f`)
- timestamp data (`MISSING`)

We need to reformat this dataset by pivoting the data in the test columns into a `test_assignment` and a `test_number` column. Then we need to add a timestamp column so we know when each test started.

Question 2: Write a query and table creation statement to make `final_assignments_qa` look like the `final_assignments` table. If you discovered something missing in part 1, you may fill in the value with a place holder of the appropriate data type.

To reformat the data to match `final_assignments` we used the following code:

```
SELECT
  item_id,
  test_a as test_assignment,
  'item_test_1' as test_number,
  CAST('2010-01-01 00:00:00' AS timestamp) AS dummy_test_start_date
FROM dsv1069.final_assignments_qa
UNION ALL
SELECT
  item_id,
  test_b as test_assignment,
  'item_test_2' as test_number,
  CAST('2010-01-01 00:00:00' AS timestamp) AS dummy_test_start_date
FROM dsv1069.final_assignments_qa
UNION ALL
SELECT
  item_id,
  test_c as test_assignment,
  'item_test_3' as test_number,
  CAST('2010-01-01 00:00:00' AS timestamp) AS dummy_test_start_date
FROM dsv1069.final_assignments_qa
UNION ALL
SELECT
  item_id,
  test_d as test_assignment,
  'item_test_4' as test_number,
  CAST('2010-01-01 00:00:00' AS timestamp) AS dummy_test_start_date
FROM dsv1069.final_assignments_qa
UNION ALL
SELECT
  item_id,
  test_e as test_assignment,
  'item_test_5' as test_number,
  CAST('2010-01-01 00:00:00' AS timestamp) AS dummy_test_start_date
FROM dsv1069.final_assignments_qa
UNION ALL
SELECT
  item_id,
  test_f as test_assignment,
  'item_test_6' as test_number,
  CAST('2010-01-01 00:00:00' AS timestamp) AS dummy_test_start_date
FROM dsv1069.final_assignments_qa;
```

2. Reformat the Data

	item_id	test_assignment	test_number	dummy_test_start_date
1	2512	1	item_test_1	2010-01-01 00:00:00
2	482	0	item_test_1	2010-01-01 00:00:00
3	2446	0	item_test_1	2010-01-01 00:00:00
4	1312	0	item_test_1	2010-01-01 00:00:00
...	...	...	...	...

Question 3: Use the `final_assignments` table to calculate the order binary for the 30 day window after the test assignment for `item_test_2` (You may include the day the test started)

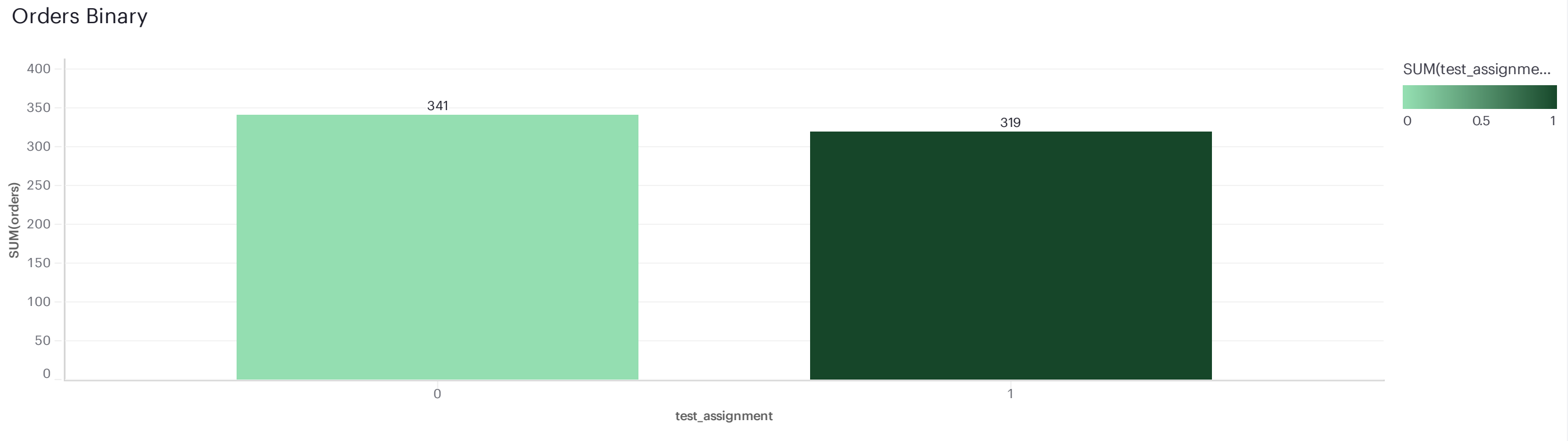
Here is the code used to compute the 30-day order binary:

```
SELECT
  order_binary.test_assignment,
  COUNT(DISTINCT order_binary.item_id) as items,
  SUM(items_after_assignment) as orders
FROM
(
  SELECT
    final_assignments.test_number,
    final_assignments.test_assignment,
    final_assignments.item_id,
    MAX(CASE
      WHEN (DATE(orders.created_at) - DATE(final_assignments.test_start_date)) BETWEEN 0 AND 30 THEN 1
      ELSE 0
    END)
    AS items_after_assignment
  FROM
    dsv1069.final_assignments
  LEFT JOIN
    dsv1069.orders
  ON
    final_assignments.item_id = orders.item_id
  WHERE
    final_assignments.test_number = 'item_test_2'
  GROUP BY
    final_assignments.item_id,
    final_assignments.test_assignment,
    final_assignments.test_number
) order_binary
GROUP BY
  order_binary.test_assignment
```

3. Compute Order Binary

	test_assignment	items	orders
1	0	1130	341
2	1	1068	319

Our results show us that for item test 2, the control group had 341 out of 1,130 items ordered, and the treatment group has 319 out of 1,068 items ordered. Below is a graph showing the two groups.



Question 4: Use the `final_assignments` table to calculate the view binary, and average views for the 30 day window after the test assignment for `item_test_2`. (You may include the day the test started)

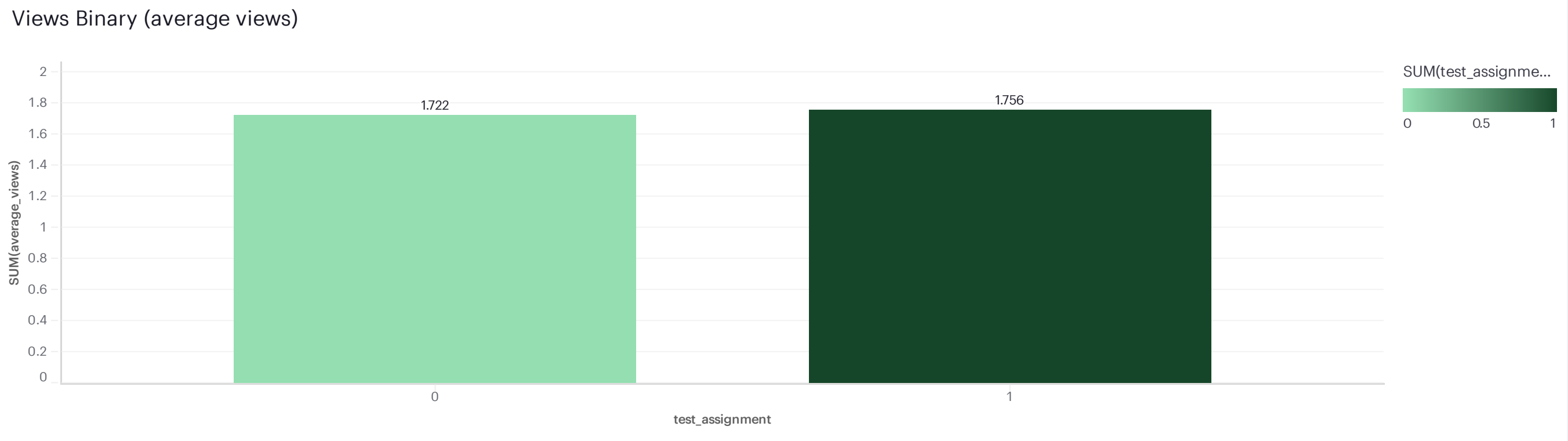
Here is the code used to compute the 30-day view binary and the average views per item:

```
SELECT
  views_binary.test_assignment,
  COUNT(DISTINCT views_binary.item_id) as items,
  SUM(views_binary.view_binary) as items_viewed,
  SUM(views_binary.total_views) as total_views,
  SUM(views_binary.total_views) / COUNT(DISTINCT views_binary.item_id) as average_views
FROM
(
  SELECT
    final_assignments.test_number,
    final_assignments.test_assignment,
    final_assignments.item_id,
    MAX(CASE
      WHEN (DATE(view_item_events.event_time) - DATE(final_assignments.test_start_date)) BETWEEN 0 AND 30 THEN 1
      ELSE 0
    END)
    AS view_binary,
    SUM(CASE
      WHEN (DATE(view_item_events.event_time) - DATE(final_assignments.test_start_date)) BETWEEN 0 AND 30 THEN 1
      ELSE 0
    END)
    AS total_views
  FROM
    dsv1069.final_assignments
  LEFT JOIN
    dsv1069.view_item_events
  ON
    final_assignments.item_id = view_item_events.item_id
  WHERE
    final_assignments.test_number = 'item_test_2'
  GROUP BY
    final_assignments.item_id,
    final_assignments.test_assignment,
    final_assignments.test_number
)
AS
  views_binary
GROUP BY
  views_binary.test_assignment
```

4. Compute View Item Metrics

	test_assignment	items	items_viewed	total_views	average_views
1	0	1130	925	1946	1.72212389381
2	1	1068	894	1875	1.75561797753

Our results show that for 925 out of 1,130 items were viewed with an average of 1.72 views per item in the control group. The treatment group had 894 out of 1,068 items viewed and an average of 1.76 views per item.



Question 5: Use the <https://thumbtack.github.io/abba/demo/abba.html> to compute the lifts in metrics and the p-values for the binary metrics (30 day order binary and 30 day view binary) using a interval 95% confidence.

--Order binary: -1% lift, p-value = 0.88. Not significant

--Views binary: 2.3% lift, p-value = 0.25. Not significant

By these metrics there is not a statistically significant difference between the control and the treatment group. This means we fail to reject the null hypothesis, meaning item test 2 was not effective at increasing orders and/or views.